



HORMONES, SEXUAL HEALTH & MISC

Melanotan-II

Melanotan-II – 10 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Ac-Nle-cyclo(-Asp-His-D-Phe-Arg-Trp-Lys-)-NH2
SYNONYMS / ALIASES	MT-II; Melanotan 2; MT2; cyclic alpha-MSH analog
CAS NUMBER	121062-08-6
MOLECULAR FORMULA	C50H69N15O9
MOLECULAR WEIGHT	1024.18 g/mol
AMINO ACID SEQUENCE	Ac-Nle-cyclo[Asp-His-D-Phe-Arg-Trp-Lys]-NH2 (7 residues; N-terminal acetyl-norleucine; lactam-bridged cyclic hexapeptide via Asp-Lys side chains; C-terminal amide)
SOURCE	Synthetic (solid-phase peptide synthesis with lactam cyclization)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	10 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Lactam-bridge cyclization confers significant enzymatic stability. The tryptophan residue is light-sensitive; protect from prolonged light exposure.

STRUCTURAL NOTES

Melanotan-II is a synthetic 7-amino-acid cyclic peptide analog of alpha-melanocyte-stimulating hormone (alpha-MSH). The peptide is constrained by a lactam bridge between the side-chain carboxylate of aspartate and the epsilon-amino group of lysine, creating an intramolecular ring. The N-terminus carries an acetyl-norleucine (Nle, a non-natural methionine isostere lacking the oxidation-prone sulfur), and the C-terminus is amidated. It is classified as a non-selective melanocortin receptor agonist with activity at MC1R, MC3R, MC4R, and MC5R. Structurally, it differs from PT-141 (bremelanotide) only in its C-terminal capping group (amide vs free carboxylate).

SELECTED PEER-REVIEWED REFERENCES

1. Al-Obeidi F. et al. - *Design of a new class of superpotent cyclic alpha-melanotropins based on quenched dynamic simulations* - Journal of the American Chemical Society, 1989 - DOI: 10.1021/ja00191a018
2. Hadley M.E., Dorr R.T. - *Melanocortin peptide therapeutics: historical milestones, clinical studies and commercialization* - Peptides, 2006 - DOI: 10.1016/j.peptides.2005.10.027
3. Dorr R.T. et al. - *Increased eumelanin expression and tanning is induced by a superpotent melanotropin [Nle4-D-Phe7]-alpha-MSH in humans* - Photochemistry and Photobiology, 2004 - DOI: 10.1562/0031-8655(2004)079<0496:IEEATI>2.0.CO;2

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.